Grounded coupler design | EC 165 / 250 / 160 / 260 MHz

Revised after mandatory dressed-idle calibration - 29 October 2026. Hard OFF limit remains below 10 GHz. All frequencies are cyclic frequencies (H/h), capacitances are fF. The earlier large integrated chi mismatch was an uncompensated-control diagnostic; it is superseded as the final readout result.

Item	Result
Three-mode exchange OFF	9.980174 GHz; modest-basis residual convergence: True
Five-mode exchange OFF	9.800131 GHz before resonator integration
Calibrated seven-mode OFF	9.902329 GHz; physical 0100-0010 null at reported basis. 1 Hz certification remains unresolved.
Earlier seven-mode CZ	119.930 ns; raw 99.7952%; postselected 99.9986%. PRIOR controls; not recalibrated after idle retuning.
Calibrated integrated readout	Both pairs pass nominal constraints and complete independent +/-1% EJ sign grids. A mismatch 0.0767%, B mismatch 0.6975%.

Confirmed input constraints

Input	Value
Loaded rail EC/h	A1 165; A2 250; B1 160; B2 260 MHz - immutable at every circuit stage
Idle operating-frequency guides	A 4.61 GHz; B 5.39 GHz. Individual bare settings compensate dressing; final dressed means are reported separately.
Isolated-pair logical splitting	A 150 MHz; B 90 MHz at the design reference; fit once and freeze hardware
Gate scope	About 100 ns static CZ; A stationary, B moved; B20 physical auxiliary (0020). Minimum-S1 toy selection owns logical source.
Coupler bounds	Loaded EC 100-600 MHz; ON >7 GHz; ON < OFF <10 GHz
Loading	Per coupler: each rail <10%; prefer <7%, minimize combined loading. Coupler own self-capacitance >=55% of effective budget. Four-branch aggregate reconstructed separately.
Topology	Floating rails; C2 directly grounded, C1 active. Actual A22-C1, C1-B11, A22-B11 capacitors. Equal pad-ground caps within each rail transmon; self caps compensate loaded EC. No unspecified parasitics.
Readout	Independent A/B designs first; nominal 50 ohm; logical-state mean readout targets 5.990/6.010 GHz. EJ/EC >45 exception. Mandatory individual-EJ idle calibration after integration.
Numerical targets	Residual couplings/ZZ: 1 Hz absolute; gate errors and survival: 1e-4 absolute probability. Finite audits stop with explicit nonconvergence.

1 | Toy selection and fresh three-body search

Quantity	Value
Selected ON channel	Logical 10 to B20
Stationary A / nominal moved B	4.610000 / 4.830473 GHz
Common intended dressed coupling	5 MHz; nominal resonant cycle 100 ns
Toy S1 / S2	0.00343080 / 0.00573366
Maximum unintended |g/detuning|	0.060798 (<0.2)
Toy calibrated time / raw / postselected	102.942 ns / 99.3951% / 99.9917%

Logical 0/1 are the lower/upper isolated-pair one-excitation modes. Pair vacuum is separate. The physical B20 label does not determine the logical source. All four logical sources were compared at common 5 MHz using the established full-cosine toy score basis.

Three-body search

Fresh independent Ca/Cb ratios from 0.125 to 8, three initial EC values and interior OFF seeds were used. The signed exchange was solved for the actual direct capacitor, giving 42 verified seed families; 11 independent continuous optimizations then minimized rail loading while retaining the OFF branch and ON-strength seed.

Parameter	Three-mode solution
A22-C1 / C1-B11 / direct A22-B11	10.151256 / 12.738629 / 0.946167 fF
Loaded coupler EC	414.346891 MHz
ON / verified retuned OFF	7.000100 / 9.980174 GHz
Reduced 11-to-02 ON strength seed	8.836696 MHz (not a logical gate certificate)
Rail-to-C loading A2 / B1	2.4143% / 1.8339%
Coupler own fraction	55.0010%

OFF nulls physical A2/B1 01-10, with C in its ground reference and virtual coupler paths included. The larger reduced ON matrix element compensates pair-hybridization charge overlaps when targeting 5 MHz in the selected logical channel. The search minimum is near the allowed upper OFF boundary; it is a local best found design, not a global proof.

2 | Five-body refinement and capacitances

A1/B2 are restored without resonators. Positive self capacitances and intra-pair links reproduce the fixed loaded ECs and A150/B90 idle splittings. The first 9.98015 GHz five-body design shifted outside 10 GHz after readout integration; it was rejected at validation. Returning to five-body design with a 9.8 GHz search ceiling supplied integration margin without relaxing any user constraint.

Actual physical capacitor	Five-mode fF	Integrated seven-mode fF
A22 - C1	6.564568	6.564568
C1 - B11	14.372491	14.372491
A22 - B11	0.786207	0.786207
A12 - A21	13.668593	21.849062
B12 - B21	6.517997	6.609151

Self capacitor	Five-mode fF	Seven-mode fF
A1 across pads (C1-ground for C)	55.832396	101.258654
A2 across pads (C1-ground for C)	34.071822	4.218413
C across pads (C1-ground for C)	23.660171	25.549591
B1 across pads (C1-ground for C)	56.409294	103.593851
B2 across pads (C1-ground for C)	35.747691	8.682536

Pad-ground capacitor (each pad)	Five-mode fF	Seven-mode fF
A1	117.395329	23.479066
A2	77.480917	131.717559
B1	121.063933	24.212787
B2	74.500882	126.651499

Loaded coupler EC = 450.284990 MHz. Maximum compensated relative EC error in five-mode design: 1.68e-16. Coupler capacitors are unchanged during readout integration; one-time self-cap and intra-pair compensation is then frozen. C2 is exactly ground, not a large capacitor approximation.

3 | Loading and OFF residuals

Budget quantity	A2	B1	Coupler
Five-mode complete-branch loading	2.2761%	2.1739%	44.9990%
Five-mode own / total effective C	93.9745%	96.5946%	55.0010%
Seven-mode own / total effective C	90.4445%	95.5695%	59.3932%
Seven-mode rail-to-C loading	2.0459%	2.3848%	40.6068%

Own capacitance is across-pad self C plus the series combination of the two pad-ground capacitors (C1-ground self C for the grounded coupler). Ceff = e^2/(2h EC). Loading is 1 - EC_loaded/EC_without_links, holding self caps fixed. The complete coupler branch includes the direct rail capacitor; the rail-to-C-only counterfactual removes the two C links. These are different reported budgets, not interchangeable percentages.

Shared rail	Four physical branches	C-link loading	Own fraction	EC MHz
A2	fixed hardware	6.6663%	88.7128%	236.0024
A2	shared pair EC compensated	7.0617%	88.0434%	250.0000
B1	fixed hardware	6.7129%	91.4879%	151.5412
B1	shared pair EC compensated	7.0876%	91.0128%	160.0000

Four-branch results reconstruct the nodal circuit with four physical coupler/opposite-pair branches attached to the same central rail. They are not four times the single-branch percentage. The compensated row is a separate static capacitance budget, not a four-neighbor dynamic gate simulation.

Modes	OFF GHz	Physical g Hz	01L-10L Hz	00L-11L Hz	ZZ Hz
Five	9.800131	7.88e-07	832.24	820.37	-111.08
Seven	9.902329	8.52e-07	852.18	840.34	7.50

Coupler EJ/h at OFF: five modes 29.295865 GHz; seven modes 29.880200 GHz. These are the bare coupler f01 coordinates, not dressed normal-mode eigenfrequencies.

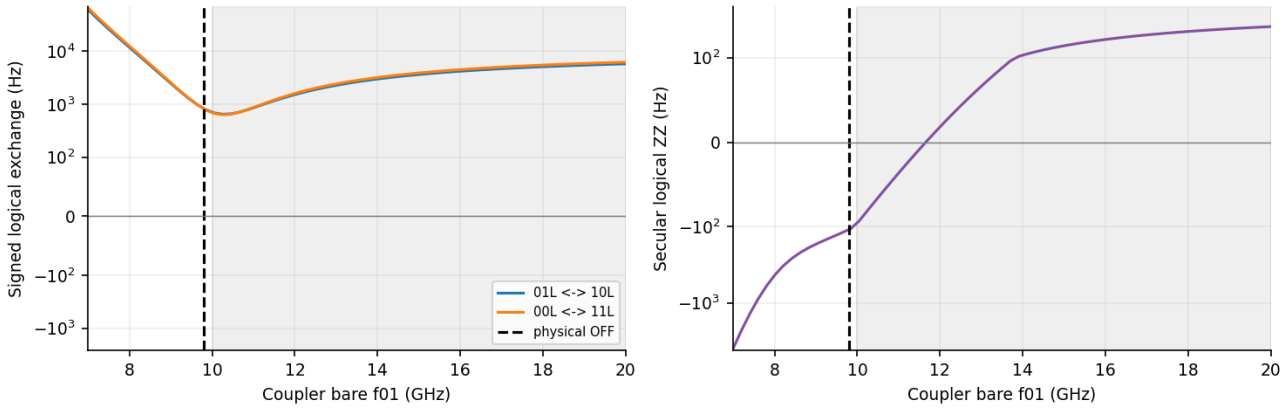
Connected physical-state detuning at OFF: five modes -755.5376 MHz; seven modes -783.5500 MHz. Logical 01L-10L detuning in seven modes is -52.5468 MHz.

Five/seven-mode OFF temporarily nulls physical 0100-0010 after canonical projection onto all four physical single-excitation references. Logical exchanges and secular ZZ remain residual diagnostics. This temporary physical-versus-logical choice remains an open workflow question. Seven-mode values above are basis-dependent estimates, not 1 Hz-certified residuals.

4 | Exchange detunings and diagnostic sweep

Connected states at seven-mode idle OFF	g (kHz)	Detuning (MHz)	|g / detuning|
01 <-> B20	-239.6825	-624.0762	0.000384
00 <-> B20	-231.7416	-717.5767	0.000323
11 <-> B20	139.1831	-478.0290	0.000291
00 <-> B02	-113.5820	-587.9270	0.000193
10 <-> B20	96.1147	-571.5295	0.000168
10 <-> B02	75.5464	-441.8797	0.000171
01 <-> B02	67.2709	-494.4264	0.000136
11 <-> B02	-47.7774	-348.3792	0.000137

Small exchange magnitudes should be assessed alongside detunings. Detunings are diagonal differences in the reported canonical/local-energy frame. Secular ZZ uses tracked logical eigenbranches: (E11-E10-E01+E00)/h.



Five-mode sweep uses local5/pair12/C5, ncut30 and frozen capacitors/idle controls; it illustrates shape rather than Hz-level pointwise convergence. Grey shading above 10 GHz is diagnostic only. The dashed line marks the retuned five-mode physical null. Signed logarithmic axes show both signs and small residuals.

5 | Prior static CZ benchmarks - unchanged controls

These gate results are retained from BEFORE the new integrated idle calibration. They are not a recalibrated gate using the newly adjusted stationary A rail. The revised idle cell records those new stationary settings for future ON calibration; no new gate fidelity is claimed in this revision.

Quantity	Five modes	Seven modes
ON coupler f01 (GHz)	7.050000	7.168900
Moved B bare f01 (GHz)	4.871278	4.862037
Duration (ns)	110.162	119.930
Conditional-phase error (rad)	8.78e-12	2.13e-09
Raw CZ fidelity	99.5145%	99.7952%
Postselected score	99.9990%	99.9986%
Mean logical survival	99.5155%	99.7967%
Worst-input nonlogical population	0.7902%	0.3934%
Actual-bias max unintended ratio	0.065893	0.089909
Fixed-control gate precision	<1e-4 between tested bases	True

Logical input	Five-mode nonlogical population	Seven-mode nonlogical population
00	0.4686%	0.1843%
01	0.7902%	0.3934%
10	0.3103%	0.1884%
11	0.3688%	0.0472%

The conditional phase is calibrated to pi before improving return. Geometry, loaded ECs and charge coefficients remain frozen; coupler EJ, moved B-pair EJ/frequency and duration may vary. A remains fixed. Starts lie within 200 MHz of the preceding solution. The finite duration search is 70-130 ns, around the requested 100 ns target; this is a local numerical search, not a global optimum proof.

All scores use consistent static ON dressed references, retaining CZ auxiliaries. There is no sudden idle-to-ON jump or ramp in this benchmark. Raw and postselected values refer to the SAME operating point. Postselected score = raw score / average logical survival, matching the toy convention; it is a survival-normalized benchmark, not a separately optimized gate. No decoherence, readout drive or pulse-shape error is included.

6 | Independent and integrated readout

Each standalone design contains only its own rail pair plus one resonator. Each uses two equal actual inner-pad-to-resonator links. The isolated-pair target is A150/B90; dressed splitting acceptance is A140-160/B80-100 MHz. No coupler or opposite rail was present during link fitting.

Readout component	A pair	B pair
Each actual rail-resonator link (fF)	9.337256	4.527794
Bare resonator LC f (GHz)	6.081051	6.056678
Resonator self C (fF)	523.445508	525.551904
Resonator L (nH)	1.308614	1.313880
Standalone mean logical readout target	5.990 GHz	6.010 GHz
Standalone all workflow constraints	[]	[]

Constraint	Required
Chi mismatch	|chi+ - chi-| / |(chi+ + chi-)/2| <1%
Readout strength	Absolute signed mean chi >=1 MHz
Dispersivity / H / overlap	Worst |g/detuning| <0.1; H >=0.95; reference overlap >=0.95
Transmon limits	Loaded EC 150-400 MHz; EJ/h <25 GHz; EJ/EC >45
Frequency / stability	Dressed operating mean below dressed resonator; chi signs stable on independent +/-1% EJ1/EJ2, 5x5 grid
Physical implementation	Equal readout links; exact nodal inverse; fixed loaded EC through self-cap compensation

Coupler-resonator dispersivity at the prior seven-mode ON bias

Resonator	Coupler transition	g (MHz)	Detuning (MHz)	|g / detuning|
r1	0-1	0.3213	1181.923	0.0002719
r1	1-2	0.4368	644.333	0.0006779
r1	2-3	0.5073	60.226	0.008424
r2	0-1	16.8197	1160.888	0.01449
r2	1-2	22.8642	623.298	0.03668
r2	2-3	26.5547	39.191	0.6776

These local charge-matrix estimates use the prior benchmark ON bias. They are distinct from the full multilevel worst dispersive ratio used to validate readout.

6b | Calibrated integrated idle readout

Exchange OFF does not remove unequal self-energy shifts of A2 and B1. Individual rail EJs are now calibrated with all geometry and loaded ECs fixed; the common coupler OFF root is then rechecked. Both rails are evaluated jointly at the resulting settings. Near resonance is assessed by the existing readout H-score and chi constraints, not by imposing equal bare frequencies.

Transmon	Original bare GHz	Calibrated bare GHz	Calibrated EJ/h GHz
A1	4.613060962	4.612856563	17.339380
A2	4.613323398	4.624675459	11.953997
B1	5.392300206	5.427281304	24.432179
B2	5.391125368	5.390259268	15.423145

Common OFF: 9.853409358 -> 9.902328515 GHz. Capacitors, resonator L, loaded ECs and all charge coefficients remain unchanged. Only rail EJs and coupler OFF EJ change.

The unequal bare settings compensate unequal interaction-induced shifts. They are not the final pair eigenfrequencies: each rail still has distinct lower/upper logical transitions, with the splitting reported below. Equal bare frequencies do not establish dressed resonance, and exchange cancellation does not remove local frequency dressing.

Comparison	A chi mismatch	B chi mismatch
Standalone readout	0.990%	0.990%
Integrated, uncompensated diagnostic	11.132%	27.709%
Integrated, calibrated final centers	0.0767%	0.6975%

The large uncompensated mismatch therefore does not establish an intrinsic inability to read out the coupled circuit. The required individual-EJ calibration restores both nominal readout constraints and sign stability. This calibration is now mandatory in the workflow; historical retuned results must not be compared with unretuned new controls as a hardware comparison.

6c | Readout at calibrated idle centers

Rail A at the saved nominal idle center

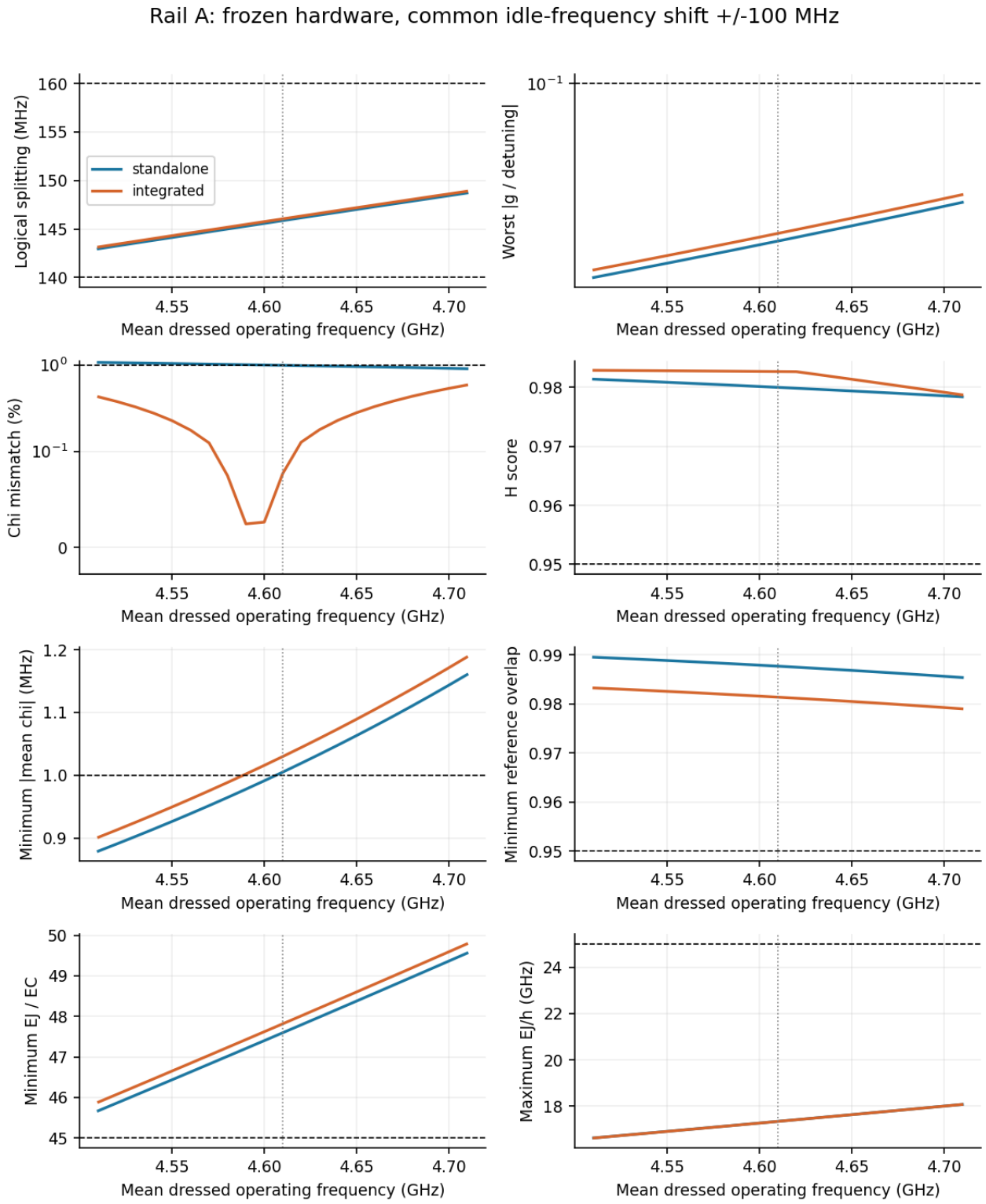
Metric	Standalone	Calibrated integrated
Mean dressed operation (GHz)	4.61	4.61001
Logical splitting min (MHz)	145.867	146.047
Worst dispersive ratio	0.0695464	0.0707847
Chi mismatch (%)	0.99	0.0767174
H score minimum	0.979989	0.982667
Minimum |mean chi| (MHz)	1.005	1.0298
Minimum overlap	0.987685	0.981372
Constraint failures	none	none

Rail B at the saved nominal idle center

Metric	Standalone	Calibrated integrated
Mean dressed operation (GHz)	5.39	5.38999
Logical splitting min (MHz)	87.2733	93.5005
Worst dispersive ratio	0.0789341	0.0891257
Chi mismatch (%)	0.99	0.697539
H score minimum	0.971009	0.950206
Minimum |mean chi| (MHz)	1.005	1.12784
Minimum overlap	0.983322	0.976143
Constraint failures	none	none

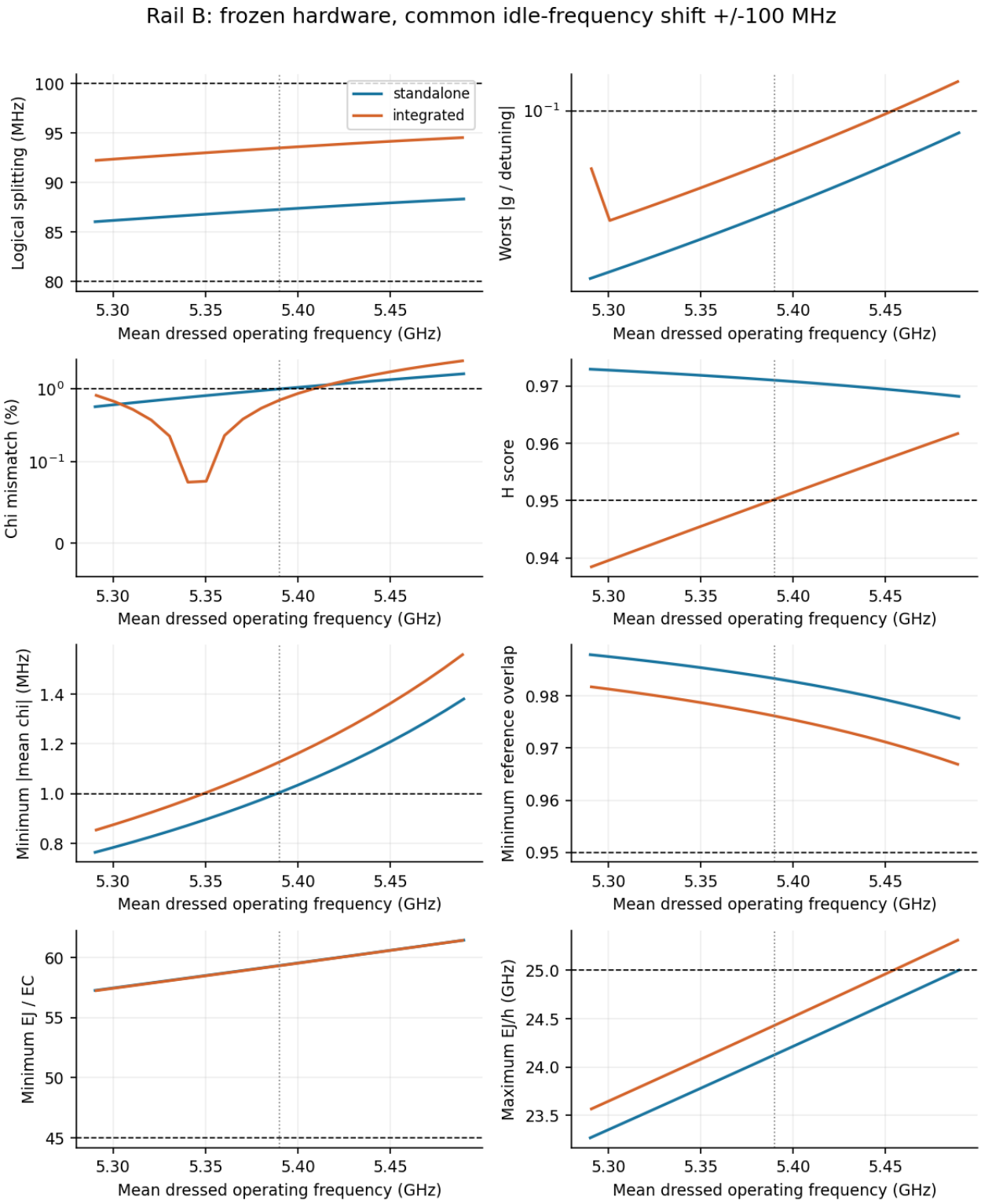
Both calibrated centers pass the complete independent +/-1% EJ1/EJ2, 5x5 sign grids, including reference-overlap reliability, in the recorded adaptive probe bases. Integrated values are worst over both spectator logical states. Sign grids are not repeated at each frequency-map point; nominal trace acceptance away from the centers is not a fresh sign certificate.

7A | Readout constraints over +/-100 MHz



Each trace varies both selected bare transmon frequencies together over +/-100 MHz around its own calibrated center. Their differential setting, all capacitors, resonator L, opposite rail EJs and coupler OFF EJ are fixed. The x coordinate is the resulting mean dressed operating frequency for each trace. Black dashed lines are actual thresholds; the grey line marks the intended operating-frequency guide. There is no new narrow frequency bound and no per-point retuning.

7B | Readout constraints over +/-100 MHz



Each trace varies both selected bare transmon frequencies together over +/-100 MHz around its own calibrated center. Their differential setting, all capacitors, resonator L, opposite rail EJs and coupler OFF EJ are fixed. The x coordinate is the resulting mean dressed operating frequency for each trace. Black dashed lines are actual thresholds; the grey line marks the intended operating-frequency guide. There is no new narrow frequency bound and no per-point retuning.

8 | Numerical checks, timing and reproducibility

OFF check at identical controls	Physical exchange change	Logical exchange change	ZZ change
Five, last two tested bases	0.199 Hz	0.121 Hz	0.024 Hz
Seven, search to modest validation	277.567 Hz	135.108 Hz	32.806 Hz

Five-mode OFF checks stop when physical exchange, both logical exchanges and secular ZZ agree within 1 Hz. The seven-mode audit table above belongs to the earlier unretuned reference; it did not meet 1 Hz. Current OFF residuals are freshly evaluated at calibrated controls with local6/pair12/C5 and three photon levels, but no new 1 Hz convergence certificate is claimed. Calibrated nominal readout and sign-stability checks pass. Full isolation and a recalibrated gate under the revised idle controls remain separate validation tasks.

Gate fixed-control checks compare raw fidelity, postselected score, mean survival and per-input leakage. Seven-mode maximum tested changes: {'raw_cz_fidelity': 4.212416759963311e-06, 'postselected_cz_fidelity': 1.0231657732173716e-07, 'logical_survival': 4.110366472409588e-06, 'worst_input_leakage_change': 1.3755454171704429e-05}. Photon levels remain three; a separate photon-cutoff convergence certificate is not supplied.

9 | Stage timings and saved records

Stage / retained attempt	Measured elapsed seconds
toy selection and calibration seconds	2.094
three body search seconds	12.460
five body design seconds	2.407
five body off validation and sweep seconds	10.774
five body gate calibration seconds	28.953
seven body integration and off seconds	46.370
Three-body fixed-control audit	2.099
Independent readout A	7.011
Standalone/integrated map A	268.075
Independent readout B	13.755
Standalone/integrated map B	268.671
Prior seven-mode gate recalibration	543.149
Dressed-idle calibration, final refinement	63.972
Calibrated idle sign-grid A	37.131
Calibrated readout map A	225.571
Calibrated idle sign-grid B	36.893
Calibrated readout map B	225.714
Rejected near-10GHz five-mode attempt (design + audit + gate)	52.444

Stage durations use perf_counter, including numerical search and verification in that stage. The interrupted/restarted seven-mode gate stage instead uses OS process-creation UTC to final endpoint time, plus its measured extra audit. Readout maps and the final gate job ran concurrently, so their durations must not be summed as total elapsed wall time. Preparation, code inspection, plotting/layout and the earlier superseded 12 GHz run are outside these numerical stage timings. The first failed seven-mode root scan took about 6.1 s process wall time. An earlier unseeded exploratory gate trial was discarded and is outside the reported final calibration timer; it is not a gate result.

Updated records: idle-retuned/final/result.json, calibrated-cell.json, post-retune-off.json, sign-A/B.json and map-A/B.json. Original cell.json, seven-off.json and map-A/B.json remain uncompensated diagnostics; seven-gate.json remains the prior-control benchmark. Hardware reconstruction and loaded ECs are unchanged. Final idle-calibration timing excludes discarded preparatory attempts; all shown map/sign jobs are timed independently and may overlap. Workflow and channel policy remain the authorities.